

# TAYLOR INVENTION

Thermally Controlled Plant Nursery Assembly — Technical Reference Manual

## 1. Patent & Administrative Information

|                           |                                  |                           |                                      |
|---------------------------|----------------------------------|---------------------------|--------------------------------------|
| Patent Publication Number | US 2024/0407307 A1               | Patent Application Number | 18/206,891                           |
| Publication Date          | December 12, 2024                | Filing Date               | June 7, 2023                         |
| Inventor / Applicant      | Gaylon Taylor<br>(Rosenberg, TX) | CPC Classifications       | A01G 9/14, A01G 9/246,<br>A01G 9/247 |

## 2. System Architecture & Assembly Specifications

The assembly comprises a housing enclosing an interior space configured with thermal and resource delivery sub-systems:

### A. Thermal Control System (TCS)

The thermal control system utilizes a refrigerant cycle to manage micro-climate temperatures:

- Evaporator (46):** Integrates custom cooling coils directly wrapped or coupled to the exterior tube (40) of each plant container. It absorbs heat from the root zone.
- Compressor (48) & Inverter (49):** An inverter-driven compressor dynamically adjusts power consumption and speed to modulate cooling cycles.
- Condenser (50) & Expansion Valve (52):** A heat-rejection condenser mounted to the housing exterior releases absorbed energy.

### B. Closed-Loop Water Supply System

Features a continuous filtration and recirculation loop:

- Overhead Sprinklers (62):** Emits misty water downwardly over containers to hydrate soil.
- Recycling Reservoir (64):** A central catchment area below the perforated shelf (28).
- Recirculation Pump (66) & Filter (68):** Draws water through a physical filter to purge particulates, feeding it back to the sprinkler system.
- Heater System (75) & Temp Sensor (79):** Controls water temperatures via heat pump compressor (77) to avoid thermal shock to roots.

## 3. Key Component Legend

| Ref No. | Component Name   | Ref No. | Component Name |
|---------|------------------|---------|----------------|
| 12      | Housing Assembly | 14      | Interior Space |

| Ref No.      | Component Name                       | Ref No.      | Component Name                                |
|--------------|--------------------------------------|--------------|-----------------------------------------------|
| 16 / 18      | Bottom Wall / Top Wall               | 28           | Perforated Support Shelf                      |
| 32           | Container Units                      | 38 / 45      | Aeration Apertures / Mesh Bottom              |
| 40 / 42      | Exterior Tube / Interior Bucket      | 44           | Thermal Control System                        |
| 46 / 48      | Evaporator Coils / Compressor        | 54 / 56      | Circuit Board / Actuator (Display)            |
| 58           | Circulation Fans                     | 60 / 62      | Water Supply / Sprinkler Jets                 |
| 64 / 66      | Reservoir Tank / Recirculation Pump  | 70           | UV Sanitizer Array                            |
| 72 / 76 / 80 | Power Supply / Gen Port / Solar Port | 75 / 77 / 79 | Heater System / Heat Compressor / Temp Sensor |

## 4. Biosecurity & Power Resilience

- **Sanitization (70):** Pathogen prevention is accomplished through UV emitters alongside customizable fluorescent and blue spectrum grow lights to maximize photosynthetically active radiation (PAR).
- **Power Interface (72):** Includes a robust, redundant input architecture. Users can seamlessly connect standard grid power (AC), portable generators (via Generator Port 76), or modular solar arrays (via Solar Panel Port 80) for full off-grid operational resilience.